

- Q.24 How does a flywheel store energy? (CO2)
 Q.25 Why is a flywheel used in engines? (CO2)
 Q.26 Define a cam and follower mechanism. (CO3)
 Q.27 What are the main functions of a cam? (CO3)
 Q.28 What is meant by constrained motion? (CO3)
 Q.29 What is meant by fluctuation of energy? (CO4)
 Q.30 What is the difference between a flywheel and a governor? (CO4)
 Q.31 Write the advantages of flywheel. (CO4)
 Q.32 Write the short note on automatic transmission. (Co5)
 Q.33 Write purpose of brake. (CO5)
 Q.34 Write the functions of clutch. (CO6)
 Q.35 Write any five application of balancing. (CO7)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Define cam and followers. Explain types of cam and followers. (CO1)
 Q.37 Explain the construction and working of single plate clutch with the help of neat sketch. (CO4)
 Q.38 Explain vibration and types of vibration. (CO7)

Note: Course Outcome (CO) mentioned in the question paper is for official purpose only.

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Roll No.

4th Sem / T & D, Prod.

Sub : Basics of Mechanical Engineering

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Thermodynamics mainly deals with. (CO1)
 a) Motion of bodies
 b) Heat and work interaction
 c) Structure of atom
 d) Electrical circuits
- Q.2 The unit of entropy is SI system is. (CO1)
 a) J b) J/kg
 c) J/k d) W
- Q.3 Which of the following is an intensive property? (CO1)
 a) Mass b) Volume
 c) Temperature d) energy
- Q.4 The main function of a flywheel is to: (CO2)
 a) Increase speed b) Store energy
 c) Reduce friction d) Increase torque

- Q.5 The time during which the follower remains at rest is called: (CO3)
- a) Lift b) Dwell
c) Return d) Stroke
- Q.6 Which of the following is used to transmit power between two parallel shafts? (CO3)
- a) Bevel gear b) Worm gear
c) Spur gear d) Cross belt
- Q.7 The velocity ratio in a simple belt drive depends on: (CO4)
- a) Length of belt b) Diameter of pulleys
c) Width of belt d) Material of belt
- Q.8 which transmission device gives exact velocity ratio? (CO4)
- a) Belt drive b) Robe drive
c) Chain drive d) Fiction wheel
- Q.9 Which type of clutch is commonly used in automobiles? (CO5)
- a) Cone clutch b) Single plate clutch
c) Centrifugal clutch d) Dog clutch
- Q.10 The logarithmic decrement is used to measure: (CO5)
- a) Frequency b) Amplitude
c) Damping d) Speed

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Zeroth law of thermodynamics deals with _____. (CO1)
- Q.12 A system that can exchange energy but not matter with surroundings is called _____. (CO1)
- Q.13 Property which depends on mass is called _____. (CO1)
- Q.14 The energy stored in a flywheel depends upon _____. (CO2)
- Q.15 Flywheel is generally mounted on _____. (CO2)
- Q.16 The base circle of a cam is _____. (CO2)
- Q.17 The main function of a key in power transmission is to: (CO3)
- Q.18 Which gear is used to transmit power between parallel shafts? (CO3)
- Q.19 The main function of a clutch is to _____. (CO5)
- Q.20 The main function of a brake is to _____. (CO6)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What are thermodynamic properties? (CO1)
- Q.22 State the Zeroth Law of Thermodynamcis. (CO1)
- Q.23 What is thermal equilibrium? (CO1)

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